Mumbai Rainfall Forecasting

PURUSHOTHAM SESHADRI
BOSTON INSTITUTE OF ANALYTICS

Problem Statement

IMPORTANCE OF ACCURATE FORECASTING

Mumbai's water management authority requires accurate rainfall forecasts to:

- Optimize reservoir storage
- Improve water allocation strategies
- Prepare for drought and heavy rainfall
- Reduce operational risk and costs
- Accurate monthly rainfall prediction is critical for sustainable resource planning.

Dataset Overview

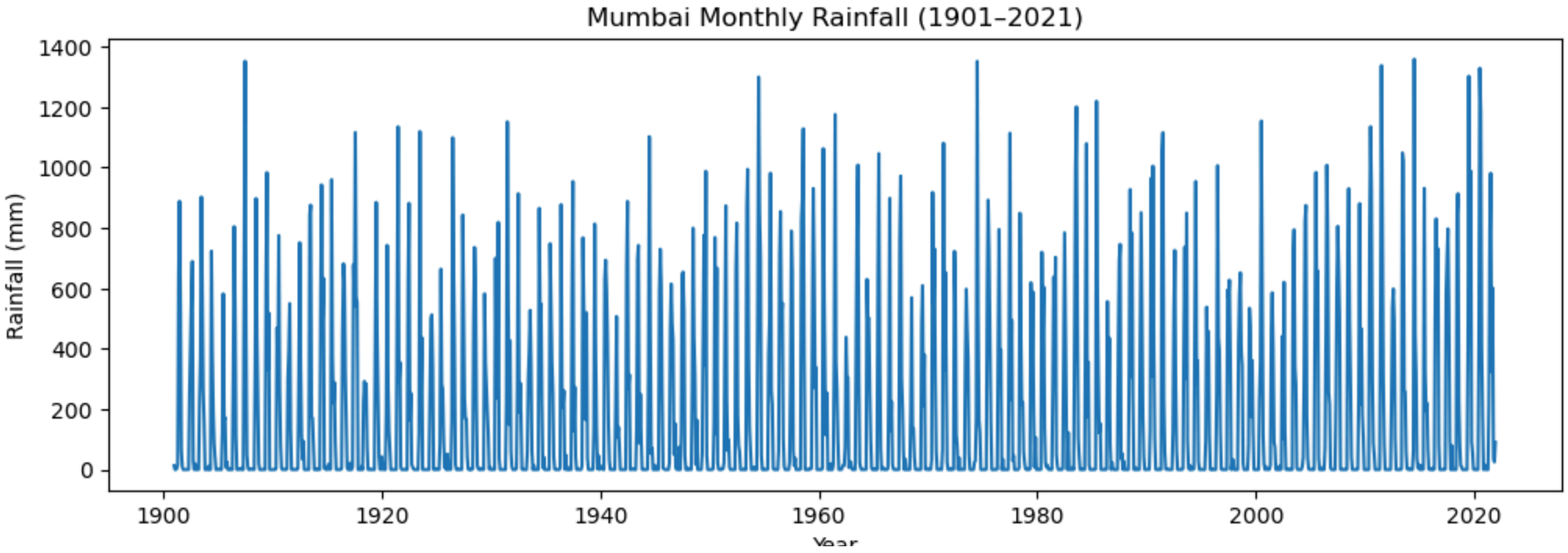
SOURCE AND CHARACTERISTICS

- Historical Monthly Rainfall Data (1901–2021)
- Location: Mumbai, India
- Frequency: Monthly
- Total Observations: 121 Years

The dataset shows strong seasonal monsoon patterns.

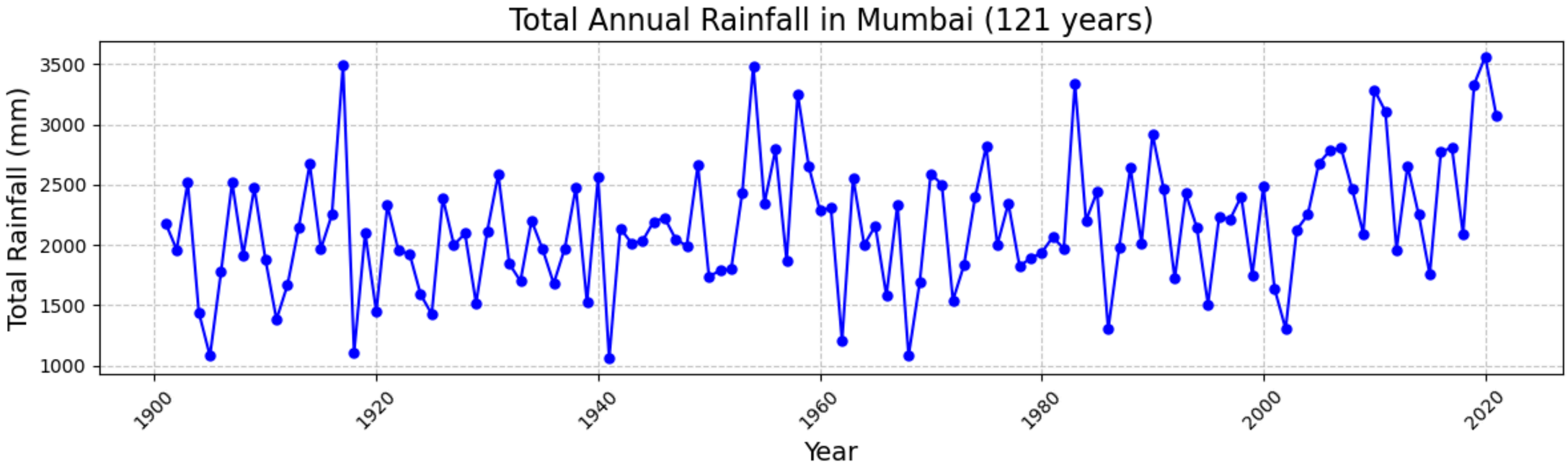
Exploratory Data Analysis of Rainfall Patterns

- Mumbai rainfall exhibits a strong and consistent seasonal monsoon pattern, with peak rainfall occurring between June and September each year.
- There is significant year-to-year variability in total rainfall, indicating fluctuating monsoon intensity.
- Several extreme rainfall spikes are observed, highlighting the occurrence of heavy monsoon events.
- No clear long-term upward or downward trend is visible, suggesting rainfall remains largely cyclical rather than progressively increasing or decreasing.
- The dominant seasonal structure makes the dataset well-suited for time-series forecasting models such as Holt-Winters and SARIMA.



Exploratory Data Analysis of Rainfall Patterns

- Mumbai rainfall exhibits a strong and consistent seasonal monsoon pattern, with peak rainfall occurring between June and September each year.
- There is significant year-to-year variability in total rainfall, indicating fluctuating monsoon intensity.
- Several extreme rainfall spikes are observed, highlighting the occurrence of heavy monsoon events.
- No clear long-term upward or downward trend is visible, suggesting rainfall remains largely cyclical rather than progressively increasing or decreasing.
- The dominant seasonal structure makes the dataset well-suited for time-series forecasting models such as Holt-Winters and SARIMA.



Models Implemented

HOLT-WINTERS

The **Holt-Winters model** utilizes exponential smoothing to effectively capture seasonal patterns, making it ideal for rainfall forecasting.

SARIMA

SARIMA integrates autoregressive and moving average components, effectively modeling seasonal trends in time series data for rainfall predictions.

RANDOM FOREST

Random Forest Regression employs ensemble learning techniques, enhancing prediction accuracy by aggregating results from multiple decision trees for rainfall forecasting.

LSTM

The LSTM model employs recurrent neural network architecture with memory cells to effectively capture long-term seasonal dependencies and nonlinear patterns in rainfall time-series data, enhancing predictive capability.

Evaluation Metrics

MAE

MAE, or **Mean Absolute Error**, measures the average magnitude of errors in predictions, providing a clear insight into overall model performance.

RMSE

RMSE, or **Root Mean Squared Error**, highlights prediction inaccuracies by penalizing larger errors, making it crucial for assessing model reliability.

R^2 SCORE

The **R^2 Score** indicates the proportion of variance in the dependent variable explained by the model, showcasing its predictive strength and effectiveness.

Model Comparison Results

Model	MAE	RMSE	R^2 Score
Holt-Winters	**96.38**	**183.29**	**0.75**
SARIMA	101.21	185.65	0.74
Random Forest	100.98	191.1	0.73
LSTM	132.98	214.26	0.66

Model Comparison Results

HOLT-WINTERS

The Holt-Winters model showcased the best overall performance, achieving the lowest MAE and RMSE values among all tested models.

RANDOM FOREST

Random Forest regression showed reasonable results, yet its performance was the lowest, highlighting the challenges of complex seasonal patterns.

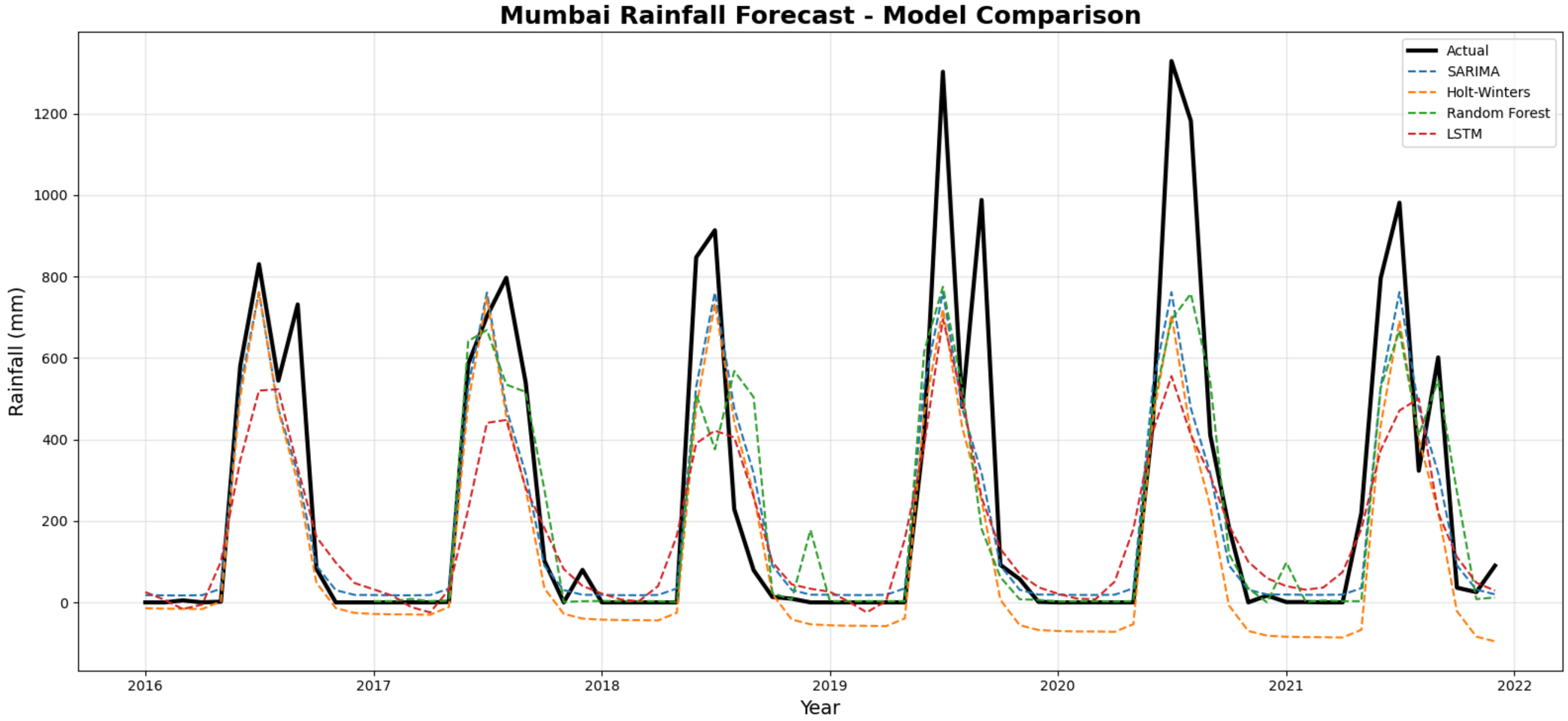
SARIMA

SARIMA performed well, but its slightly higher MAE and RMSE compared to Holt-Winters indicated less accuracy in rainfall predictions.

LSTM

The LSTM model showed reasonable predictive capability, but its comparatively lower performance highlights the challenges of modeling highly seasonal rainfall data with limited training samples.

Visualization of Model Comparison



Best Model: Holt-Winters

PERFORMANCE

The Holt-Winters model exhibited the **lowest MAE and RMSE**, demonstrating superior accuracy in capturing Mumbai's rainfall patterns over time.

SEASONALITY

Its ability to effectively capture **seasonal variations** in rainfall makes it particularly suited for Mumbai's distinct monsoon climate.

RELIABILITY

By achieving the **highest R^2 Score**, this model not only enhances prediction reliability but also supports informed decision-making for water resource management.

Business Impact of Forecasting

ENHANCING WATER MANAGEMENT

- Improved reservoir storage planning
- Preparation for drought and heavy rainfall
- Optimized water distribution
- Reduced operational risk and costs
- The selected model provides reliable seasonal forecasting, supporting strategic decision-making for sustainable water resource management in Mumbai.

Limitations and Future Work

CHALLENGES AND IMPROVEMENTS

- Rainfall is naturally volatile and complex.
- Current model excludes external climate variables.
- Future improvements may incorporate:
  - Temperature data
  - ENSO index
  - Humidity data
  - Hybrid modeling approaches

Conclusion

Multiple forecasting models were evaluated to predict Mumbai rainfall.
The Holt-Winters model performed best in capturing seasonal patterns and achieved the highest predictive accuracy.
This forecasting system provides a practical and reliable decision-support tool for improved water resource management.